

Kathy Wen

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Professional Summary

I am a current DPhil candidate in Engineering Science at the University of Oxford, specialising in offshore engineering. My research focuses on soil-structure interaction through experimental and theoretical investigations. With prior experience in construction and design within the infrastructure sector, I have contributed to multiple projects in transportation, residential, and energy sectors across Australia. Having extensive industry exposure, I recognise the importance of a strong grasp of fundamental engineering and physical principles in developing optimised solutions for complex problems. Consequently, I am interested in opportunities that contribute to understanding of essential theoretical engineering concepts and techniques.

Education

Expected graduation 2023	University of Oxford Doctor of Philosophy in Engineering Science Research topic: Application of small-strain stiffness for monopile design	Oxford
Feb 2018	University of New South Wales Bachelor of Civil Engineering (Honours class 1)	Sydney

Publications

Wen, K. Z., Buckley, R. M., and Byrne, B.W.: Obtaining representative stiffness degradation parameters from the resonant column, Proceedings of the 9th International Offshore Site Investigation and Geotechnics Conference, 2023 (Accepted)

Dehtyriov, D., Wen, K.Z., and Byrne, B.W.: The impact of wind turbine aerodynamics on offshore foundation design, Journal of Physics: Conference series, TORQUE, 2024 (Under review)

Experience

Oct 2022 – present	University of Oxford Lead Tutor – Soil Mechanics and Geotechnics - Lead tutor for two engineering courses, B10 Soil mechanics and C6 Geotechnics - Prepared and presented tutorials based on lecture material - Set and marked mid-year paper (collections paper) for B10 course - Provided detailed feedback to students based on performance in tutorials and homework sheets	Oxford
Oct 2022 – present	Regent's Park College Academic tutor - Created and delivered interactive seminars over structured 2-month programme. - Seminars focused on academic reading, writing and presenting - Marked and provided feedback to students group presentations	Oxford
Nov 2021 – present	St Peter's College Casual academic - Created STEM related lecture style academic sessions targeted at a high school students - Delivered interactive presentations to various groups sizes (ranging 2 to 40 students) with content related to engineering and statistics	Oxford

Oct 2019 – Present	University of Oxford DPhil Candidate - Planned and supervised ground investigations for material sampling - Calibration and installation of an advanced laboratory device for soil characterisation - Planned and executed rigorous laboratory testing regime to investigate research hypothesis - Perform data processing and interpretation to investigate research hypothesis - Implementation of numerical algorithms to perform geotechnical analysis	Oxford
July 2018 – Oct 2019	SMEC Pty Ltd. Geotechnical Engineer - Planned, procured, and supervised ground investigations for construction projects - Analysed site investigation data and interpreted laboratory test results to complete detailed characterisation for several projects - Undertook various geotechnical analysis using numerical modelling techniques and theoretical solutions - Prepared factual and design engineering reports to provide site-specific geotechnical recommendations	Sydney
Jan 2018 – July 2018	Lendlease Graduate Engineer - Undertook high level estimates of refurbishment projects used to establish budgets for the project lifetime - Oversaw daily construction activities to monitor programme and budget - Subcontract negotiation and commercial close out of projects	Sydney
Nov 2016 – Jan 2018	Seymour Whyte Pty Ltd Undergraduate Engineer - Managed and delivered several civil construction packages including, earthworks, drainage, kerb and gutter, foundations, and pavement - Determined project programme of works, including sequencing of construction activities for optimal efficiency - Conducted routine quality audits, verifying progression against specifications, and initiating corrective actions to remedy concerns - Avoided construction delays by maintaining thorough site inspections and efficient communication with team members	Sydney
July 2017 – Jan 2018	University of New South Wales Geotechnical Demonstrator - Prepared and presented a range of geotechnical topics including, 'Applied pile design', 'Shallow foundations', and 'Soil mechanics' - Assisted up to 30 students with geotechnical queries and assignments - Marked and provided feedback on students' submitted work and exams	Sydney

Skills

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| • Geotechnical design and analysis | • Matlab |
| • Site investigation and <i>in-situ</i> testing | • AutoCAD |
| • Geotechnical laboratory testing and interpretation | • Latex |
| • Education and communication | • Bently Systems software – primarily PLAXIS-2D and 3D |
| • Project management | • GeoStudio software |
| | • Microsoft 365 products |

Volunteering

Present	Oxford mutual aid	Oxford
2021	Jacari	Oxford
2015	Engineers without Borders	Sydney
2015	St Vincent De Paul	Sydney
2011-2014	Red Shield Appeal	Sydney
